Past paper 2018. MATH-201

(i) Basis of Column space: $\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 2 \end{bmatrix} \Rightarrow R \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{matrix} R_2-3R_1 \\ R_3-R_1 \end{matrix} \Rightarrow R \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & -1 & 3 \end{bmatrix} (-1)R$$

$$R \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_3+R_2 \text{ reduced echelon form.} \quad R = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

leading 1 wale columns

C_1 and C_3 from original matrix

$e_1 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$ and $e_2 = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$ form basis for column space of given matrix

Basis of Row space $= \{ r_1 = [1\ 2\ 2\ -1],\ r_2 = [0\ 0\ 1\ -3] \}$

Dimension of row space = Rank of A

$\Rightarrow$ Nullity(A) + Rank(A) = n

$N = 3 - 2 = 1$

(ii) If A is nonsingular inverse is $\begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$, find A

$A^{-1} = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$ $\quad (A^{-1})^{-1} = \dfrac{\text{Adj } A^{-1}}{|A^{-1}|}$ $\quad |A^{-1}| = 2 - 4 = -2$

$\text{Adj } A^{-1} = \begin{bmatrix} 1 & -1 \\ -4 & 2 \end{bmatrix}$ $\quad (A^{-1})^{-1} = \begin{bmatrix} -1/2 & 1/2 \\ 2 & -1 \end{bmatrix} = A$

(iii) $\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{bmatrix}$ into reduced echelon form.

$$R \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 0 & 7 & -7 & 6 \end{bmatrix} \begin{matrix} R_2-2R_1 \\ R_3-3R_1 \end{matrix} \quad R \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 1 & -4/3 & 4/3 \\ 0 & 7 & -7 & 6 \end{bmatrix} \tfrac{1}{3}R_2 \approx R \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 1 & -4/3 & 4/3 \\ 0 & 0 & 7/3 & -10/3 \end{bmatrix} R_3-7R_2$$

$$R \begin{bmatrix} 1 & -2 & 3 & -1 \\ 0 & 1 & -4/3 & 4/3 \\ 0 & 0 & 1 & \frac{-10}{7} \end{bmatrix} \tfrac{3}{7}R_3 \quad R \begin{bmatrix} 1 & -2 & 0 & \frac{23}{7} \\ 0 & 1 & 0 & -4/7 \\ 0 & 0 & 1 & -10/7 \end{bmatrix} \begin{matrix} R_2+\frac{4}{3}R_3 \\ R_1-3R_3 \end{matrix} \quad R \begin{bmatrix} 1 & 0 & 0 & 15/7 \\ 0 & 1 & 0 & -4/7 \\ 0 & 0 & 1 & -10/7 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 & 1 & 0 \\ -2 & -2 & 2 & 0 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/2 & 0 \\ 0 & 2 & -4 & 2 & 0 & 1 \end{array}\right] = \left[\begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/2 & 0 \\ 0 & 0 & -4 & 2 & -1 & 1 \end{array}\right] = \left[\begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/2 & 0 \\ 0 & 0 & 1 & -1/2 & -1/4 & 1/4 \end{array}\right] = \left[\begin{array}{ccc} -1/2 & \cdots \\ 0 & 1 & \cdots \\ 0 & 0 & 1 \end{array}\right.$$

(iv) $\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$

$$= \begin{vmatrix} (b+c)^2-a^2 & 0 & a^2 \\ 0 & (c+a)^2-b^2 & b^2 \\ c^2-(a+b)^2 & c^2-(a+b)^2 & (a+b)^2 \end{vmatrix} \begin{matrix} C_1-C_3 \\ C_2-C_3 \end{matrix} = \begin{vmatrix} (b+c-a)(b+c+a) & 0 & a^2 \\ 0 & (c+a-b)(c+a+b) & b^2 \\ (c-a-b)(c+a+b) & (c-a-b)(c+a+b) & (a+b)^2 \end{vmatrix}$$

Taking $(a+b+c)$ common from C_1, C_2 both

$$= (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ c-a-b & c-a-b & 2ab \end{vmatrix} = (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ -2b & -2a & 2ab \end{vmatrix} R_3-(R_1+R_2)$$

$= (a+b+c)^2\{(b+c-a)(2ab(c+a-b)+2ab^2) - 0 + a^2(0+2b(c+a-b))\}$

$= (a+b+c)^2((b+c-a)(2abc+2a^2b-2ab^2+2ab^2)+2a^2b(c+a-b))$

$= (a+b+c)^2((b+c-a)(2ab)(c+a)+2a^2b(c+a-b))$

$= (a+b+c)^2\, 2ab((b+c-a)(c+a)+a(c+a-b))$

$= 2ab(a+b+c)^2(bc+c^2-ac+ab+ac-a^2+ac+a^2-ab)$

$= 2abc(a+b+c)^2(b+c+a) = 2abc(a+b+c)^3$

(v) W is subspace of V or not? $V = R^3$

$W = \{(a,b,c) \in V : a^2+b^2+c^2 \leq 1\}$

(i) $(1,0,0) \in W \Rightarrow W \neq \phi$

(ii) $w_1 = (1,0,0)$ $w_2(0,1,0) \Rightarrow \alpha w_1 + \beta w_2 = \alpha(1,0,0)+\beta(0,1,0)$

$(\alpha, \beta, 0) \notin W$ Not vector space

LONG QUESTION

3. Find A^{-1} of $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$ $A:I$ 4+6

$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$ $|A| = 1(-4+0)-2(0)-3(0-0) = -4$

$\text{Adj } A = \begin{bmatrix} a_{11} & -a_{12} & a_{13} \\ -a_{21} & a_{22} & -a_{23} \\ a_{31} & -a_{32} & a_{33} \end{bmatrix}^T$

$$\left|\begin{matrix} 3/4 & -3/4 \\ 1/2 & 0 \\ -1/4 & 1/4 \end{matrix}\right| = \left[\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}\right| \left.\begin{matrix} 1/2 & +1/4 & -3/4 \\ 0 & -1/2 & 0 \\ -1/2 & -1/4 & 1/4 \end{matrix}\right]$$

$A_{11} = \begin{vmatrix} -2 & 0 \\ -2 & 2 \end{vmatrix} = -4$ $\quad A_{21} = \begin{vmatrix} 2 & -3 \\ -2 & 2 \end{vmatrix} = 4-6 = -2$ $\quad A_{31} = \begin{vmatrix} 2 & -3 \\ -2 & 0 \end{vmatrix} = -6$

$A_{12} = \begin{vmatrix} 0 & 0 \\ -2 & 2 \end{vmatrix} = 0$ $\quad A_{22} = \begin{vmatrix} 1 & -3 \\ -2 & 2 \end{vmatrix} = 2-6 = -4$ $\quad A_{32} = \begin{vmatrix} 1 & -3 \\ 0 & 0 \end{vmatrix} = 0$

$A_{13} = \begin{vmatrix} 0 & -2 \\ -2 & -2 \end{vmatrix} = -4$ $\quad A_{23} = \begin{vmatrix} 1 & 2 \\ -2 & -2 \end{vmatrix} = -2+4 = 2$ $\quad A_{33} = \begin{vmatrix} 1 & 2 \\ 0 & -2 \end{vmatrix} = -2$

$$\text{Adj } A = \begin{bmatrix} -4 & -0 & -4 \\ -(-2) & -4 & -2 \\ -6 & -(0) & -2 \end{bmatrix}^T = \begin{bmatrix} -4 & 0 & -4 \\ 2 & -4 & -2 \\ -6 & 0 & -2 \end{bmatrix}^T$$

$$= \begin{bmatrix} -4 & 2 & -6 \\ 0 & -4 & 0 \\ -4 & -2 & -2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & -1/2 & \text{[illegible]} \\ 0 & 1 & 0 \\ 1 & 1/2 & 1/2 \end{bmatrix}$$

4. $v_1(1,-2,3)$, $v_2 = (5,6,-1)$, $v_3 = (3,2,1)$ linearly independent?

$av_1 + bv_2 + cv_3 = (0,0,0)$ where $a,b,c \in R$

$a(1,-2,3) + b(5,6,-1) + c(3,2,1) = (0,0,0)$

$(a+5b+3c, -2a+6b+2c, 3a-b+c) = (0,0,0)$

$a+5b+3c=0 \quad -2a+6b+2c=0 \quad 3a-b+c=0$

$$A = \begin{bmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{bmatrix} \quad B \begin{bmatrix} 1 & 5 & 3 \\ 0 & 16 & 8 \\ 0 & -16 & -8 \end{bmatrix} \begin{matrix} \\ R_2+2R_1 \\ R_3-3R_1 \end{matrix} \quad B \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 1/2 \\ 0 & -16 & -8 \end{bmatrix} \tfrac{1}{16}R_2$$

$$B \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix} R_3+16R_1$$

Not linearly independent

$1(6+2)-5(-2-6)+3(2-18)$

$8-5(-8)+3(-16)$

$8+40-48$

$48-48=0$

As $|A|=0$ So it is linearly dependent

5. $\{(1,2,-1),(0,3,1),(1,-5,3)\}$ is basis of R^3?

$a(1,2,-1)+b(0,3,1)+c(1,-5,3)=(x,y,z)$

$(a+c,\ 2a+3b-5c,\ -a+b+3c)=(x,y,z)$

$$\begin{bmatrix}1&0&1\\2&3&-5\\-1&1&3\end{bmatrix} \underset{\sim}{R} \begin{bmatrix}1&0&1\\0&3&-3\\0&1&4\end{bmatrix} \begin{matrix}\\R_2-2R_1\\R_3+R_1\end{matrix} \quad \underset{\sim}{R} \begin{bmatrix}1&0&1\\0&1&-1\\0&1&4\end{bmatrix}$$

$$\underset{\sim}{R} \begin{bmatrix}1&0&1\\0&1&-1\\0&0&5\end{bmatrix}$$

$c=0=a=b$ These vectors are basis of R^3

6. $\begin{bmatrix}1&2\\2&1\end{bmatrix}$ find $P^{-1}AP$

$A-\lambda I=\begin{bmatrix}1-\lambda&2\\2&1-\lambda\end{bmatrix}=(1-\lambda)(1-\lambda)-4=1-2\lambda+\lambda^2-4=\lambda^2-2\lambda-3$

Characteristic Polynomial

$\lambda^2-3\lambda+\lambda-3=\lambda(\lambda-3)+1(\lambda-3)=(\lambda-3)(\lambda+1)$ $\quad\lambda=3,-1$ Eigen values

$A-3I=\begin{bmatrix}1-3&2\\2&1-3\end{bmatrix}=\begin{bmatrix}-2&2\\2&-2\end{bmatrix}=\begin{bmatrix}1&-1\\2&-2\end{bmatrix}=\begin{bmatrix}1&-1\\0&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}x_1-x_2\\0\end{bmatrix}$ $\quad x_1=x_2$

$\begin{bmatrix}x_1\\x_1\end{bmatrix}=x_1\begin{bmatrix}1\\1\end{bmatrix}$ $\quad[1\ \ 1]^T$ Eigen vector. We normalize it

$\left[\frac{1}{\sqrt2}\ \ \frac{1}{\sqrt2}\right]^T$

$A+I=\begin{bmatrix}2&2\\2&2\end{bmatrix}=\begin{bmatrix}1&1\\0&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}x_1+x_2\\0\end{bmatrix}$ $\quad x_1=x_2 \Rightarrow \begin{bmatrix}x_1\\-x_1\end{bmatrix}=x_1\begin{bmatrix}1\\-1\end{bmatrix}$

$[1\ \ -1]^T$ $\quad\left[\frac{1}{\sqrt2}\ \ \frac{-1}{\sqrt2}\right]^T$

$P=\begin{bmatrix}\frac{1}{\sqrt2}&\frac{1}{\sqrt2}\\\frac{1}{\sqrt2}&\frac{-1}{\sqrt2}\end{bmatrix}^T$ $\quad P=P^T=P^{-1}=\begin{bmatrix}\frac{1}{\sqrt2}&\frac{1}{\sqrt2}\\\frac{1}{\sqrt2}&\frac{-1}{\sqrt2}\end{bmatrix}$

$P^{-1}AP=\frac{1}{2}\begin{bmatrix}1&1\\1&-1\end{bmatrix}\begin{bmatrix}1&2\\2&1\end{bmatrix}\begin{bmatrix}1&1\\1&-1\end{bmatrix}=\frac{1}{2}\begin{bmatrix}1&1\\1&-1\end{bmatrix}\begin{bmatrix}3&-1\\3&1\end{bmatrix}=\frac{1}{2}\begin{bmatrix}6&0\\0&-2\end{bmatrix}=\begin{bmatrix}3&0\\0&-\end{bmatrix}$

7. $x+y+7z=-7$
$2x+3y+17z=-16$
$x+2y+(a^2+1)z=3a$

Determine values of a for which system of linear equation has no solution, exactly one solution and infinitely many solution.

$$\left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 2 & 3 & 17 & -16 \\ 1 & 2 & a^2+1 & 3a \end{array}\right]$$

$$\underset{\sim}{R}\left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 1 & a^2-6 & 3a+7 \end{array}\right] \begin{array}{l} R_2-2R_1 \\ R_3-R_1 \end{array} \qquad \underset{\sim}{R}\left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & a^2-9 & 3a+9 \end{array}\right] R_3-R_2$$

No solutions if

$a^2-9=0$ $\quad 3a+9\neq 0$

$a=\pm 3$

One solution if

$a^2-9\neq 0 \Rightarrow a\neq \pm 3$ and $3a+9$ is real number

Many solution if

$a=\pm 3$ $\quad a=-3 \Rightarrow [0\ 0\ 0\ |\ 0]$

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i) Decide yes...... If A is 2×2 matrix such that $A^2=0$, A must be null matrix

No

Counter example: $A=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow A^2=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$A^2=0$ but A is not null matrix

In this case, A is nilpotent matrix, meaning some power of A (in this case, square) is zero matrix but A itself is not zero matrix

(ii) $v=(2,-5,3)$ as linear combination $u_1(1,-3,2)$, $u_2=(2,-4,-1)$, $u_3(1,-5,7)$

$v=au_1+bu_2+cu_3$

$(2,-5,3)=a(1,-3,2)+b(2,-4,-1)+c(1,-5,7)$

$(2,-5,3) = (a+2b+c, -3a-4b-5c, 2a-b+7c)$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ -3 & -4 & -5 & -5 \\ 2 & -1 & 7 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 1 \\ 0 & -5 & 5 & -1 \end{bmatrix} \begin{matrix} R_2+3R_1 \\ R_3-2R_1 \end{matrix}$$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 1/2 \\ 0 & -5 & 5 & -1 \end{bmatrix} \tfrac{1}{2}R_2 \qquad \left[\begin{array}{ccc:c} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 1/2 \\ 0 & 0 & 0 & 3/2 \end{array}\right] R_3+5R_1$$

Rank of A ≠ Rank of Ab

(iii) $V = \{f : f : R \to R\}$ $W = \{f \in V : f(1) = 0\}$ Subspace or not?

$V = \{f : R \to R\}$ $W = \{f \in V : f(1) = 0\}$

(i) $0(1) = 0 \to 0 \in W$

(ii) $f, g \in W$ $f(1) = 0$ $g(1) = 0$

$(f+g)(1) = f(1) + g(1) = 0 + 0 = 0 \Rightarrow f+g \in W$

(iii) $\alpha \in R$

$(\alpha f)(1) = \alpha f(x) = \alpha(0) = 0 \Rightarrow \alpha f \in W$

W is Subspace of V

(iv) $$\begin{vmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix} = (a-1)^3(a+3)$$

$$\begin{vmatrix} a+3 & 1 & 1 & 1 \\ a+3 & a & 1 & 1 \\ a+3 & 1 & a & 1 \\ a+3 & 1 & 1 & a \end{vmatrix} C_1+(C_2+C_3+C_4) = (a+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix}$$

$$= (a+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & a-1 & 0 & 0 \\ 0 & 0 & a-1 & 0 \\ 0 & 0 & 0 & a-1 \end{vmatrix} = (a+3)(a-1)^3$$

2) If A and B are invertible matrices of same order AB is invertible then show $(AB)^{-1} = B^{-1}A^{-1}$

$(AB)^{-1} = B^{-1}A^{-1}$

$\therefore (AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}$ Associative law

$= AIA^{-1} = I$ $\quad \because BB^{-1} = I,\ AA^{-1} = I$

Similarly,

$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B$

$= B^{-1}IB = I$

So, $(AB)(B^{-1}A^{-1}) = (B^{-1}A^{-1})(AB)$

$(AB)^{-1} = B^{-1}A^{-1}$

This proof assumes that A, B are invertible matrices of same order, same number of rows and columns.

3. $x_1 - 2x_2 + 3x_3 = 3$

$2x_1 + x_2 - x_3 = 9$

$-3x_1 - 4x_2 + 5x_3 = -15$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 2 & 1 & -1 & 9 \\ -3 & -4 & 5 & -15 \end{array}\right] \approx \left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & -7 & 3 \\ 0 & -10 & 14 & -6 \end{array}\right] \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 + 3R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 1 & -7/5 & 3/5 \\ 0 & -10 & 14 & -6 \end{array}\right] \frac{1}{5}R_2 = \left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 1 & -7/5 & 3/5 \\ 0 & 0 & 0 & 0 \end{array}\right] R_3 + 10R_2$$

Rank of $A \neq$ Rank of A_b

4. $T(x_1, x_2, x_3) = (-x_3, x_1, x_1 + x_3)$

let $u_1 = (x_1, x_2, x_3)$ $\quad u_2 = (y_1, y_2, y_3)$

$Tu_1 = T(-x_3, x_1, x_1 + x_3)$

for $N(T) \Rightarrow Tu = 0$

$x_3 = 0 \quad x_1 = 0 \quad x_1 + x_3 = 0$

$0 = 0$ satisfied

$N(T) = \{(x_1, x_2, x_3) \in R^3 : (0, x_2, 0)\}$

It is not one-to-one

5. $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ $A^2 - 4A - 5I = ?$

$$A^2 = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix}$$

$$4A = \begin{bmatrix} 4 & 8 & 4 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} \qquad 5I = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

= [illegible]

6. $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ $P^{-1}AP = ?$ ① Done

7. $M\begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}$ $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $C\begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$

$$M = C_1A + C_2B + C_3C$$

$$\begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} = C_1\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + C_2\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + C_3\begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$$

$4 = C_1 + C_2 + C_3$

$7 = C_1 + 2C_2 + C_3$

$7 = C_1 + 3C_2 + 4C_3$

$9 = C_1 + 4C_2 + 5C_3$

$$\begin{bmatrix} 1 & 1 & 1 & 4 \\ 1 & 2 & 1 & 7 \\ 1 & 3 & 4 & 7 \\ 1 & 4 & 5 & 9 \end{bmatrix}$$

$$\underset{R}{\sim}\begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 2 & 3 & 3 \\ 0 & 3 & 4 & 5 \end{bmatrix}\begin{matrix} \\ R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{matrix} \quad \underset{R}{\sim}\begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 4 & -10 \end{bmatrix}\begin{matrix} \\ R_3 - 2R_2 \\ R_4 - 3R_2 \\ \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 4 & -10 \end{bmatrix} \frac{1}{3}R_3 \quad \underset{\sim}{R} \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -6 \end{bmatrix} R_4 - 4R_3$$

$C_3 = -1$, $C_2 = 3$, $C_1 + C_2 + C_3 = 4 \Rightarrow C_1 = 4 + 1 - 3 = 2$

Verification:-

$$\begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} = 2\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 3\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 5 \\ 5 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}$$

Satisfied

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(i) $f(t) = \cos t$, $g(t) = \sin t$, $h(t) = t$ are L.I

$a\cos t + b\sin t + ct = 0$ $a, b, c \in R$

Diff w.r.t $-a\sin t + b\cos t + c = 0$

Again diff. w.r.t $-a\cos t - b\sin t = 0$

Put $t = 0$ $a = 0$ $b = 0$ $c = 0$ L.I

(ii)
$$\begin{vmatrix} b+c & b & c \\ c & c+a & a \\ b & a & a+b \end{vmatrix} = 2a(b^2 + c^2)$$

$= (b+c)\{(c+a)(a+b) - a^2\} - b\{ac + bc - ab\} + c\{ac - bc - ab\}$

$= (b+c)\{ac + bc + a^2 + ab - a^2\} - abc - b^2c + ab^2 + ac^2 - bc^2 - abc$

$= abc + b^2c + ab^2 + ac^2 + bc^2 + abc - 2abc - b^2c + ab^2 + ac^2 - bc^2$

$= 2ab^2 + 2ac^2$

$= 2a(b^2 + c^2)$

(iii)
$$\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$$ is involutary.

$$A^2 = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

So Given matrix is involutary

iv) $V = R^2$ and W is set of I^{st} quad of R^2

1. $W \neq \phi$

$(0,0),(1,1),(2,2) \in R^2$

2. $u \in W$ $v \in W$ $u+v \notin W$

as $(1,1)+(2,-3) = (3,-2)$ which $\notin W$ as it lies in 2^{nd} quad

3. $a \in R$ $au \notin W$ as

$-1(1,1) = (-1,-1) \notin W$

Not subspace of V

(v) $\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 0 \end{bmatrix}$ basis of Column space = ?

$\underset{\sim}{R} \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{matrix} \\ R_2-3R_1 \\ R_3-R_1 \end{matrix}$ $\underset{\sim}{R} \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & -1 & 1 \end{bmatrix} (-1)R_2$

$\underset{\sim}{R} \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & -2 \end{bmatrix} R_3+R_2$ $\underset{\sim}{R} \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
$c_1 \quad c_3 \quad c_4$

$\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$ Basis of Column space

3. $x+y+2z=9,\ 2x+4y-3z=1,\ 3x+6y-5z=0$

Gaussian elimination

$A = \begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$

$\underset{\sim}{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{bmatrix} \begin{matrix} \\ R_2-2R_1 \\ R_3-3R_1 \end{matrix}$ $\underset{\sim}{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 3 & -11 & -27 \end{bmatrix} \frac{1}{2}R_2$

$\underset{\sim}{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & -1/2 & -3/2 \end{bmatrix} R_3-3R_2$ $\underset{\sim}{R} \begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1 & 3 \end{bmatrix} -2R_3$

$\boxed{z=3}$

$y - \frac{7}{2}z = \frac{-17}{2} \Rightarrow y = \frac{-17}{2} + \frac{21}{2} = \frac{4}{2} = \boxed{2 = y}$

$x + y + 2z = 9 \Rightarrow x + 2 + 6 = 9 \Rightarrow x = 9 - 8 \Rightarrow \boxed{x = 1}$

$x = 1,\ y = 2,\ z = 3$

4. $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$ find inverse. Done

5. $\begin{bmatrix} 1 & 1 & k & 1 & k \\ 0 & -1 & 1 & 2 & 0 \\ 1 & 2 & 1 & -1 & -k \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & k & 1 & k \\ 0 & 1 & -1 & -2 & 0 \\ 0 & 1 & 1-k & -2 & -2k \end{bmatrix} = \begin{bmatrix} 1 & 1 & k & 1 & k \\ 0 & 1 & -1 & -2 & 0 \\ 0 & 0 & 2-k & 0 & -2k \end{bmatrix}$

$2 - k = -2k$

$2 = -2k + k$

$\boxed{-2 = k}$

6. $v_1(1,-2,3)$ $v_2 = (5,6,-1)$, $v_3(3,2,1)$ are L.I?

$a(1,-2,3) + b(5,6,-1) + c(3,2,1) = (0,0,0)$

$(a+5b+3c,\ -2a+6b+2c,\ 3a-b+c) = (0,0,0)$

$\begin{bmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 5 & 3 \\ 0 & 16 & 8 \\ 0 & -16 & 8 \end{bmatrix} R_2 + 2R_1 = \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{bmatrix}$

L.D

7. $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

As it is upper triangular and its eigenvalues are 1, 1, 1.

$P^{-1}AP = D$

As Eigen vectors are $(1,0,0), (0,1,0), (0,0,1)$

These are L.I so $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ which means

As P is actually identity matrix

A is already diagonalizable

2020 MATH-201

(i) $v_1 = (1,-2,3)$ $v_2 = (5,6,-1)$, $v_3 = (3,2,1)$

$a(1,-2,3) + b(5,6,-1) + c(3,2,1) = (0,0,0)$

$(a+5b+3c, -2a+6b+2c, 3a-b+c) = (0,0,0)$

$\begin{pmatrix} 1 & 5 & 3 \\ -2 & 6 & 2 \\ 3 & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 5 & 3 \\ 0 & 16 & 8 \\ 0 & -16 & -8 \end{pmatrix} \sim \begin{pmatrix} 1 & 5 & 3 \\ 0 & 1 & 1 \\ 0 & -16 & -8 \end{pmatrix} \sim \begin{pmatrix} 1 & 5 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 8 \end{pmatrix}$ $c=0=a$

L.D

(ii) $(A^T - 2I)^{-1} = \begin{pmatrix} 3 & 0 \\ 1 & -1 \end{pmatrix}$ find A

$|Det| = -3$

$Adj = \begin{pmatrix} -1 & 0 \\ -1 & 3 \end{pmatrix}$

Inverse $= \begin{pmatrix} 1/3 & 0 \\ 1/3 & -1 \end{pmatrix}$

$A^T - 2I = \begin{pmatrix} 1/3 & 0 \\ 1/3 & -1/3 \end{pmatrix}$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1/3 & 0 \\ 1/3 & -1 \end{pmatrix}$

$a - 2 = 1/3$, $b = 0$, $c = 1/3$, $d - 2 = -1$

$a = 7/3$ $d = 1$

So $A = \begin{pmatrix} 7/3 & 0 \\ 1/3 & 1 \end{pmatrix}$

(iii) $T(x_1, x_2, x_3) = (-x_3, x_1, x_2 + x_3)$ Done

(iv) $V = \{f : R \to R\}$ $W = \{f \in V : f(1) = 0\}$ Done

(v) $f(t) = \tan t$, $gt = \sin t$, $ht = \cos t$ L.I?

$a \tan t + b \sin t + c \cos t = 0$

$a \frac{\sin t}{\cos t} + b \sin t + c \cos t = 0$

$a \sin t + b \sin t \cos t + c \cos^2 t = 0$

$a \sin t + b \sin t \cos t + c(1 - \sin^2 t) = 0$

$(a+b) \sin t + \sin^2 t (c - b) + c = 0$

$t = 0$ $b = 0$

$t = \pi/2$ $a + b = 0$ $a + c = 0$

$t = \pi$ $c - b = 0$

$a = b = c = 0$

3. $2x + y + z = 1$, $3x + y - 5z = 8$, $4x - y + z = 5$

$$\begin{bmatrix} 2 & 1 & 1 & 1 \\ 3 & 1 & -5 & 8 \\ 4 & -1 & 1 & 5 \end{bmatrix} = \begin{bmatrix} [illegible] & [illegible] & [illegible] & -7 \\ 3 & 1 & -5 & 8 \\ 4 & -1 & 1 & 5 \end{bmatrix} R_1 - R_2 = \begin{bmatrix} 1 & 0 & -6 & 7 \\ 3 & 1 & -5 & 8 \\ 4 & -1 & 1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -6 & 7 \\ 0 & 1 & 13 & -13 \\ 0 & -1 & 25 & 23 \end{bmatrix} \begin{matrix} R_2 - 3R_1 \\ R_3 - 4R_1 \end{matrix} = \begin{bmatrix} 1 & 0 & -6 & 7 \\ 0 & 1 & 13 & -13 \\ 0 & 0 & 38 & 10 \end{bmatrix} R_3 + R_2$$

$$\begin{bmatrix} 1 & 0 & -6 & 7 \\ 0 & 1 & 13 & -13 \\ 0 & 0 & 1 & \frac{10}{38} \end{bmatrix}$$

$\boxed{z = \frac{5}{19}}$

$y + 13\left(\frac{5}{19}\right) = -13$

$y = -13 - \frac{65}{19}$

$\boxed{y = \frac{312}{19}}$

7.

4. $\begin{vmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{vmatrix}$ Done

5. $\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} = x^2y^2$

6. $v_1(0,0,0), v_2(1,0,1), v_3(1,-1,0)\ v_4(1,0,-1)$

$a(0,0,0)+b(1,0,1)+c(1,-1,0)+d(1,0,-1)$

(i) 0

7. Basis of Null space. $\begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 0 \end{bmatrix}$

$Ax = 0$

$A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ 1 & 2 & 1 & 0 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & -2 \end{bmatrix}$

$x+2y+2z-w=0$

$0+0-z+3w=0$

$z=3w, \quad x=-2y-2z+w \quad y =$ free variable

let $y=1, w=1$

$z=3, \ y=1, \ x=-3$

$N(T) = \{(-3,1,3,1)\}$ Dim $= 1$

2021. Math 201

i) $T(-x_3, x_1, x_1+x_3) =$ One-to-one Done

ii) $W = \{(x,y) : x, y \geq 0\}$

$W \neq \phi$

$u_1 = (1, 2)$ $u_2 = (2, 3)$

$\alpha u_1 + \beta u_2 = \alpha(1,2) + \beta(2,3) = (\alpha + 2\beta, 2\alpha + 3\beta) \notin W$

$\alpha u = \alpha(1,2) \notin W$ W not subspace of R^2

iii) $\begin{pmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{pmatrix}$ Reduced echelon form

Done

(iv) Show that similar matrices have same eigenvalues

let A and B be similar matrices, there exist invertible matrix P such that $B = PAP^{-1}$

We want to show A, B have same eigenvalues

let square A is similar to B, there exist a non-singular matrix P

Consider

$|B - \lambda I| = |P^{-1}AP - \lambda I|$

$= |P^{-1}AP - \lambda P^{-1}P|$

$= |P^{-1}(A - \lambda I)P|$

$= |P^{-1}||A - \lambda I||P|$

$= \frac{1}{|P|}|A - \lambda I||P|$

$|B - \lambda I| = |A - \lambda I|$ Characteristic Equation of A = A

Hence proved

v) let u be vector in V, assume u is orthogonal to v in V.

$u \cdot v = 0$ $\forall v \in V$

$v = u \Rightarrow u \cdot u = 0$

$\|u\|^2 = 0$ square of length (magnitude) of v

Simple method

If $u \in V$ is orthogonal to every $v \in V$

$\langle u, v \rangle = 0 \quad \forall v \in W$

In particular for $u \in V$ we have

$\langle u, u \rangle = 0 \quad \because \|u\| = \sqrt{\langle u, u \rangle}$

$\|u\| = 0 \Rightarrow \|u\| = 0 \Rightarrow u = 0$

7 $\|u\| = 0 \quad u$ is zero vector

Therefore, if u is orthogonal to every $v \in V$, u must be zero vector.

This result make special sense, as if a vector is orthogonal every other vector in space, it must "pointing in a direction" that is $\perp$ to all directions which can only be zero vector.

vi) Non similar matrices have same charactristic polynomial

$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

$t^2 - 4t + 4$

$\lambda = 2, \lambda = 2$, Eigenvalues

A and B are not similar, do not have same eigenvector. A have eigenvector $[1 \ 0]$ and $[0 \ 1]$

B has eigenvector $[1 \ 0]$ $[1 \ 1]$

or $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \Rightarrow (t-1)(t-2)^2 = C.P$

1) $x + y + 7z = -7$

$2x + 3y + 17z = -16$ Done.

$x + 2y + (a^2+1)z = 3a$

ii) $\begin{bmatrix} 1 & 2 & -3 \\ 0 & -2 & 0 \\ -2 & -2 & 2 \end{bmatrix}$ Done

iii) $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = (2-\lambda)(6-5\lambda+\lambda^2-2) - 2(2-\lambda-1) - 1(2-3+\lambda)$

$= (2-\lambda)(\lambda^2-5\lambda+4) - 2(1-\lambda) - 1(-1+\lambda)$

$= 2\lambda^2 - 10\lambda + 8 - \lambda^3 + 5\lambda^2 - 4\lambda - 2 + 3\lambda$

$= -\lambda^3 + 7\lambda^2 - 11\lambda + 5$

$$\begin{array}{c|cccc} & 1 & -7 & 11 & -5 \\ 1 & \downarrow & 1 & -6 & 5 \\ \hline & 1 & -6 & 5 & 0 \end{array}$$

$\lambda - 1 = 0$, $\lambda^2 - 6\lambda + 5 = 0$

$\lambda^2 - 5\lambda - \lambda + 5 = 0$

$\lambda(\lambda - 5) - 1(\lambda - 5) = 0$

$(\lambda - 5)(\lambda - 1) = 0$

for $\lambda = 5$,
$$\begin{vmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 2 & -3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -2 & 1 \\ -3 & 2 & 1 \\ 1 & 2 & -3 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 1 \\ 0 & -4 & 4 \\ 0 & 4 & -4 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{vmatrix}$$

iv. $2x_1 - x_2 - x_3 = 4$ $\quad 3x_1 + 4x_2 - 2x_3 = 11$ $\quad 3x_1 - 2x_2 + 4x_3 = 11$

$x_1 - 2x_2 + x_3 = 0$ $\quad x_2 - x_3 = 0$

$x_1 - 2x_3 + x_3 = 0$

$x_1 = x_3$

$[1 \ 1 \ 1]^T$

for $\lambda = 1$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 + 2x_2 + x_3$$

$x_2 = a \quad x_3 = b$

$x_1 = -2a - b$

$$\begin{bmatrix} -2a-b \\ a \\ b \end{bmatrix} = \begin{bmatrix} -2a \\ a \\ 0 \end{bmatrix} + \begin{bmatrix} -b \\ 0 \\ b \end{bmatrix} = a\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$[-2 \ 1 \ 0]^T, [-1 \ 0 \ 1]^T$

(iv)
$$\begin{bmatrix} 2 & -1 & -1 & 4 \\ 3 & 4 & -2 & 11 \\ 3 & -2 & 4 & 11 \end{bmatrix}$$

7. $= \begin{bmatrix} 1 & -1/2 & -1/2 & 2 \\ 3 & 4 & -2 & 11 \\ 3 & -2 & 4 & 11 \end{bmatrix} \frac{1}{2}R_1$

$= \begin{bmatrix} 1 & -1/2 & -1/2 & 2 \\ 0 & 11/2 & -1/2 & 5 \\ 0 & -1/2 & 11/2 & 5 \end{bmatrix} \begin{matrix} R_2-3R_1 \\ R_3-3R_1 \end{matrix} = \begin{bmatrix} 1 & -1/2 & -1/2 & 2 \\ 0 & 1 & -1/11 & 10/11 \\ 0 & -1/2 & 11/2 & 5 \end{bmatrix} \frac{2}{11}R_2$

$= \begin{bmatrix} 1 & -1/2 & -1/2 & 2 \\ 0 & 1 & -1/11 & 10/11 \\ 0 & 0 & \frac{60}{11} & \frac{60}{11} \end{bmatrix} R_3+\frac{1}{2}R_2 = \begin{bmatrix} 1 & -1/2 & -1/2 & 2 \\ 0 & 1 & -1/11 & 10/11 \\ 0 & 0 & 1 & 1 \end{bmatrix} \frac{11}{60}R_3$

$\begin{bmatrix} 1 & -1/2 & 0 & 5/2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} R_2+\frac{1}{11}R_3 \\ R_1+\frac{1}{2}R_3 \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} R_1+\frac{1}{2}R_2$

$x_1 = 3$

(v) Find equation of subspace spanned by (1, -3, 5) (-2, 6, -10)

$a(1,-3,5)+b(-2,6,-10) = (x,y,z)$

$a-2b, -3a+6b, 5a-10b = x, y, z$

$\begin{bmatrix} 1 & -2 & x \\ -3 & 6 & y \\ 5 & -10 & z \end{bmatrix} = \begin{bmatrix} 1 & -2 & x \\ 0 & 0 & y+3x \\ 0 & 0 & z-5x \end{bmatrix} \begin{matrix} R_2+3R_1 \\ R_3-5R_1 \end{matrix}$

$x = t \quad y = -3t \quad z = 5t$ parametric equations of straight line

2022

(i) All vector space must be abelian group under multiplication.

from paper

(ii) Show that linear independent set of vectors of any vector space can be extended for basis for V.

let V be vector space and let $S = \{v_1, v_2, \ldots, v_n\}$ be linearly independent set of vectors in V

We want to show that S can be extended to basis of V.

1. If <S> spans S, S is basis of V
2. If S does not span, there exists a vector v in V that is not span of S. By linear independence of S, the set $S' = \{v_1, v_2, \ldots, v_n, v\}$ is still L.I
3. Repeat 2 step until span S' equal to V. This process must terminate since V is finite dimensional
4. The resulting set S' is basis of V extended original lin. I set S.

Since S' spans V, it is basis of V. This result is fundamental property of vector space

3. Let V be Vspace over field and $W \subseteq V$. Then W is subspace of V or not + example.

i- zero vector of V is in W $(x^2+y^2+z^2)=25$

ii- for any $u, v \in W$, $u+v \in W$

iii. for any $a \in F$, $v \in W$, $av \in W$

let $V = R^3$, $W = \{(x, y, 0) \mid x, y \in R\}$

i- $W \neq \phi$

ii- $\alpha(1,1,0) + \beta(2,1,0) = (\alpha+2\beta, \alpha+\beta, 0) \in W$ subspace of V

4. Orthogonal:

A square matrix is orthogonal if it satisfies

1. $Q^T = Q^{-1}$
2. $QQ^T = Q^TQ = I$

transpose of O.M is its inverse and multiplying matrix by its transpose result in I.M

it is often used in linear trans[f] rotations, orthogonal projection

Similar:

Two matrices A and B are similar if there exists an invertible matrix P such that $B = P^{-1}AP$

Similar matrix have properties

i) Same det, trace, eigenvalues, rank, nullity

It is used to analyze linear transformations

A is similar to B
$\Rightarrow$ $B = P^{-1}AP$
Multiply $(P^{-1})^{-1}$ from left
$B(P^{-1})^{-1}(P^{-1})^{-1}P^{-1}AP = IAP = AP$

Multiply P^{-1} from right
$(P^{-1})^{-1}BP^{-1} = APP^{-1}$
$X^{-1}BX = AI$ $X = P^{-1}$
$B = A$
Hence proved

7 5. Dependence / Independence vectors:
let $\{v_1, v_2, v_3, \ldots v_n\} \in V$ V is vector space, let $\{a_1, a_2, \ldots a_n\} \in F$ then
$a_1v_1 + a_2v_2 + \ldots + a_nv_n = 0$ $\forall a_i \in F$ $a_i \neq 0$
the V is linear dependence otherwise it is said to be L.I

6. Determinant, cofactor, minors of square square.
The det of square matrix A is scalar value denoted by det(A) or |A| if
$A = \begin{bmatrix} a & e \\ f & h \end{bmatrix} = |A| = ah - ef$
for larger matrix we can use laplace expansion or other method
The cofactor of element of square matrix A is value obtained by multiplying the minor of a_{ij} by $(-1)^{i+j}$ where
- a_{ij} is element with ith row, jth column
- It is denoted by C_{ij}

Minor:
Minor of element a_{ij} of square matrix is det of submatrix formed by removing ith row and jth column i.e. If A is 2-by-2 matrix, minor a_{11} = det of 2x2 submatrix obtained by removing first row and first column

- det $(AB^2) = 72$, det $(A^2B^2) = 144$, det (A), det(2A), det(AB
- $T(x_1, x_2, x_3) = (x_1 - x_2, x_3)$ is it linear find N(T), R(T)

As $(x_1 - x_2, x_3) = x_1(1,0) + x_2(-1,0) + x_3(0,1)$
L.I
Now $T(1,0,0) = (1,0)$ $T(0,0,1) = (0,1)$
$T(0,1,0) = (-1,0)$

$R(T) = \{(1,0),(0,1),(-1,0)\}$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$ $(-1,0)$ cast it out

$R(T) = \{(1,0),(0,1)\}$ Dim of $R(T) = 2$

for $N(T)$ $Tu = 0$

$x_1 - x_2 = 0$ $\quad x_3 = 0$

$x_1 = x_2$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$ $N(T) = \{(1,1,0)\}$ Dim of $N(T)$

3. $2x_1 - x_2 - x_3 = 4$, $3x_1 + 4x_2 - 2x_3 = 11$, $3x_1 - 2x_2 + 4x_3 = 11$ Done

4. Eigen values $\begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$

$$A - \lambda = \begin{vmatrix} 5-\lambda & 9 & 3 \\ -3 & 5-\lambda & 6 \\ -1 & -5 & -3-\lambda \end{vmatrix}$$

$(5-\lambda)((5-\lambda)(-3-\lambda)+30) - 9(9+3\lambda+6) + 3(15+5-\lambda)$

$= (5-\lambda)(-15-5\lambda+3\lambda+\lambda^2+30) - 9(15+3\lambda) + 3(2$

$= (5-\lambda)(\lambda^2-2\lambda+15) - 135 - 27\lambda + 60 - 3\lambda$

$= 5\lambda^2 - 10\lambda + 75 - \lambda^3 + 2\lambda^2 - 15\lambda - 30\lambda - 75$

$= -\lambda^3 + 7\lambda^2 - 55\lambda = 0$

$-\lambda(\lambda^2 - 7\lambda + 55) = 0$

$\lambda = 0$ $\quad \lambda^2 - 7\lambda + 55 = 0$ complex

$$A - 0 = \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$$

$$\frac{x_1}{\begin{vmatrix} 9 & 3 \\ 5 & 6 \end{vmatrix}} = \frac{x_2}{\begin{vmatrix} 5 & 3 \\ -3 & 6 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 5 & 9 \\ -3 & 5 \end{vmatrix}}$$

$$\frac{x_1}{39} = \frac{x_2}{39} = \frac{x_3}{52}$$

$$\begin{bmatrix} 39 \\ 39 \\ 52 \end{bmatrix}$$

7. iii) $M_{2\times2} = \begin{bmatrix}1 & 2\\ -1 & 0\end{bmatrix} \begin{bmatrix}2 & 3\\ -2 & 1\end{bmatrix} \begin{bmatrix}a & b\\ c & d\end{bmatrix}$

$$c_1\begin{bmatrix}1 & 2\\ -1 & 0\end{bmatrix} + c_2\begin{bmatrix}2 & 3\\ -2 & 1\end{bmatrix} + c_3\begin{bmatrix}a & b\\ c & d\end{bmatrix} = \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$c_1+2c_2+c_3a=0 \quad 2c_1+3c_2+bc_3=0 \quad -c_1-2c_2+c_3c=0$

$c_2+dc_3=0$

$$\begin{bmatrix}1 & 2 & a\\ 2 & 3 & b\\ -1 & -2 & c\\ 0 & 1 & d\end{bmatrix}$$

The last matrix has $R=2$ if

$c+a=0 \quad b+d-2a=0$

$$\begin{bmatrix}1 & 2 & a\\ 0 & -1 & b-2a\\ 0 & 0 & [illegible]\\ 0 & 0 & [illegible]\end{bmatrix}$$

The required conditions on $a, b,$

$a+c=0 \quad b+d-2a=0$

$a=b=d=1, c=-1$

$c_1=c_3=1, c_2=-1$

2022.

i) If $|A|\neq 0 \quad A^t\neq 0 \quad (A^t)^{-1}=(A^{-1})^t$

Given that,

$|A|\neq 0 \quad |A^t|\neq 0$

n is non zero integer

$(A^t)^{-1}=(A^{-1})^t$

A is nonsingular

$AA^{-1}=I$

$(AA^{-1})^t=I^t$

$(A^{-1})^tA^t=I$

Multiply $(A^t)^{-1}$

$(A^{-1})^tA^t(A^t)^{-1}=I(A^t)^{-1}$

$(A^{-1})^tI=(A^{t-1})$

$(A^{-1})^t=(A^t)^{-1}$ proved

ii) For square matrix A satisfies equation $A^2+2A+I=0$ ha

$|A|\neq 0$

Given that,

$A^2 - A + I = 0$

Multiply A^{-1} on b.s

$A^{-1}A^2 - AA^{-1} + IA^{-1} = A^{-1} \cdot 0$

$A^{-1}AA - AA^{-1} + IA^{-1} = 0$

$AI - I + IA^{-1} = 0$

$A^{-1} = -A + I$

So $|A| \neq 0$

iii) Using row operation A^{-1} if $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$

[illegible]

$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$ $R_2 + 3R_3$, $R_1 - 3R_3$

$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$ $R_1 - 2R_2$ $-14 - 26$ $6 + 10$ $3 + 6$

iv) Domain, range of $T(x_1, x_2, x_3, x_4) = (x_1, x_2)$

let $U = (x_1, x_2, x_3, x_4)$ $\quad V = y_1, y_2, y_3, y_4$

$TU = x_1, x_2$ $\quad TV = y_1, y_2$

We can see domain of T is R^4 and range of T is R^2

v) Area of Δ $(3, 3), (4, 0), (-2, -1)$

$= \frac{1}{2} \left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right]$

$= \frac{1}{2} \left| 3(0+1) + 4(-1-3) + (-2)(3-0) \right| = \frac{1}{2} \left| 3 - 16 - 6 \right| = \frac{1}{2} |-19|$

$= \frac{19}{2}$

7. i (ii) Show $u = (-2, 3, 1, 4)$, $v = (1, 2, 0, -1)$ orthogonal vectors of R^4. Also find angle b/w u and v.

u and v are orthogonal if $\langle u, v \rangle = 0$

$\langle u, v \rangle = -2 + 6 - 4 = 0$

$\langle u, v \rangle = 0$ and angle is 90

So u is orthogonal to v

Q If A is invertible iff not eigen value of A.

let A be square matrix of order n

Suppose that A is invertible i.e. A^{-1} exists

$\det A \neq 0$

$\det(A - (0)I) \neq 0$

$\det(A - \lambda I) \neq 0$ of characteristic

$\lambda = 0$ is not a root eigen value of A

equation hence $\lambda = 0$ is not " "

i Conversely suppose $\lambda = 0$ characteristic equation

λ is not root of

$\det(A - \lambda I) \neq 0$

$\det((0)I - A) \neq 0$

$\det(-A) \neq 0$

$(-1)^n \det A \neq 0$

$\det A \neq 0$

A is an invertible matrix

Prove.

2. $T(x,y,z) = (x, z, x+y)$ is matrix transformation

find $R(T)$ and $N(T)$

$T: R^3 \to R^3$

$T(1,0,0) = (1,0,1)$

$T(0,1,0) = (0,0,1)$

$T(0,0,1) = (0,1,0)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \approx \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \approx \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$R(T) = \{(1,0,1)(0,0,1)(0,1,0)\}$

Dimension of $R(T) = 3$

Now $N(T) = \{(x,y,z) \in R^3 \mid T(x,y,z) = 0\}$

$x = 0$

$z = 0$

$y = 0$

$(0\ 0\ 0)$ $N(T)$ dimension of $N(T) = 0$

4. $(1,1,-4,-3)(2,0,2,-2)(2,-1,3,2)$

$a(1,1,-4,3) + b(2,0,2,-2) + c(2,-1,3,2)$

$(a+2b+2c, a-c, -4a+2b+3c, 3a-2b+2c)$

$a+2b+2c = 0$ $\quad$ $a-c=0$

$-4a+2b+3c=0$ $\quad$ $a=c$

$-3a-2b+2c=0$ $\quad$ $a=0$

$2b+3c=0$ $\quad$ $b=0$

$\cancel{2b}+c=0$

$4c=0$

$c=0$

its dimension is 3